MOORE TWINING ASSOCIATES, INC.

GEOTECHNICAL ENGINEERING INVESTIGATION
PROPOSED CLASSROOM PROJECT
GARLAND RANCH REGIONAL PARK
700 WEST CARMEL VALLEY ROAD
CARMEL VALLEY, CALIFORNIA

Project Number: H07802.01

For:

Monterey Peninsula Regional Park District (MPRPD)
4860 Carmel Valley Road
Carmel, California 93923

November 21, 2023

www.mooretwining.com

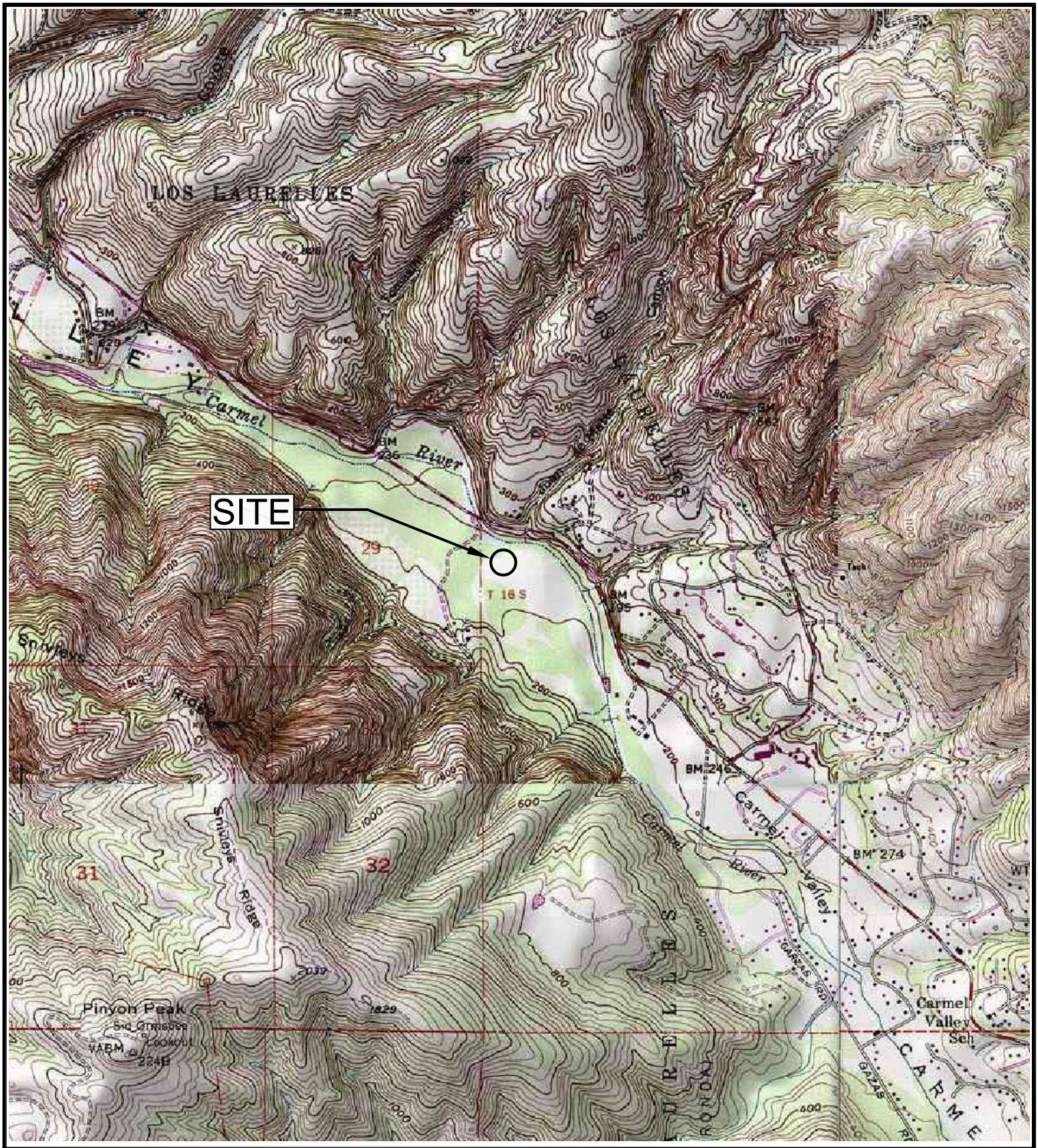
PH: 559.268.7021
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2527 Fresno Street
Fresno, CA 93721

APPENDIX A

DRAWINGS

Drawing No. 1 - Site Location Map

Drawing No. 2 - Test Boring Location Map



SOURCE: U.S.G.S. TOPOGRAPHIC MAP, 7 ½ MINUTE SERIES
SEASIDE, CALIFORNIA QUADRANGLE 1981, PHOTOREVISED 1983

0 2000
APPROXIMATE SCALE
IN FEET

SITE LOCATION MAP
PROPOSED GARLAND RANCH REGIONAL PARK
CLASSROOM PROJECT (OPEN-SIDED SHADE STRUCTURE)
700 WEST CARMEL VALLEY ROAD
CARMEL VALLEY, MONTEREY COUNTY, CALIFORNIA

FILE NO.: 07802-01-01	DATE: 11/21/2023
DRAWN BY: RM	APPROVED BY:
PROJECT NO. H07802.01	DRAWING NO. 1

MOORE TWINING ASSOCIATES, INC.



APPROXIMATE TEST BORING LOCATION

N

0 30

APPROXIMATE SCALE IN FEET

TEST BORING LOCATION MAP
PROPOSED GARLAND RANCH REGIONAL PARK
CLASSROOM PROJECT (OPEN-SIDED SHADE STRUCTURE)
700 WEST CARMEL VALLEY ROAD
CARMEL VALLEY, MONTEREY COUNTY, CALIFORNIA

FILE NO. 07802-01-02	DATE DRAWN: 11/21/2023
DRAWN BY: RM	APPROVED BY:
PROJECT NO. H07802.01	DRAWING NO. 2

MOORE TWINING ASSOCIATES, INC.

APPENDIX B

LOGS OF BORINGS

This appendix contains the final logs of borings. These logs represent our interpretation of the contents of the field logs and the results of the field and laboratory tests.

The logs and related information depict subsurface conditions only at these locations and at the particular time designated on the logs. Soil conditions at other locations may differ from conditions occurring at these test boring locations. Also, the passage of time may result in changes in the soil conditions at these test boring locations.

In addition, an explanation of the abbreviations used in the preparation of the logs and a description of the Unified Soil Classification System are provided at the end of Appendix B.

MOORE TWINING ASSOCIATES, INC.

Test Boring: B-1

Project: Proposed Garland Ranch Regional Park Classroom Project, Carmel Valley

Project Number: H07802.01

Drilled By: J.S.

Drill Type: CME 75

Auger Type: 6-5/8" O.D. Hollow Stem Auger

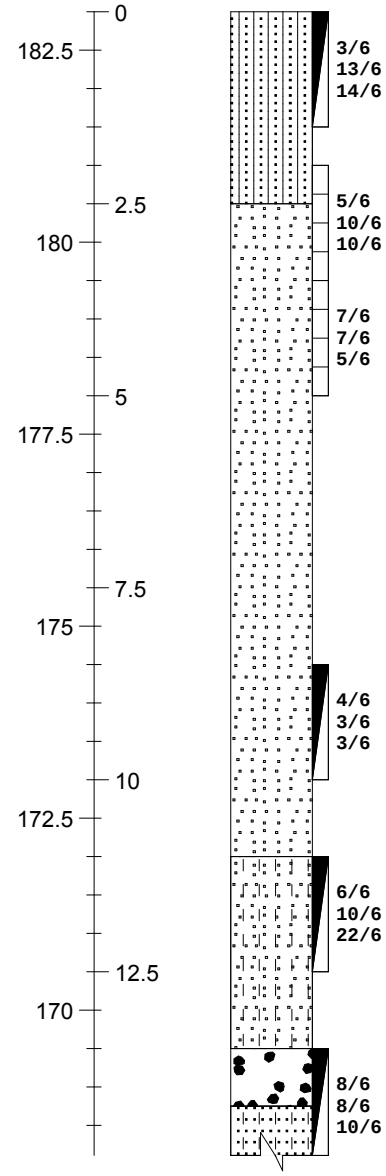
Hammer Type: 140 Pound Auto Trip

Logged By: A.H.

Date: October 19, 2023

Elevation: 183 feet AMSL

Depth to Groundwater First Encountered During Drilling: 18.2 feet*



ELEVATION/ DEPTH (feet)	USCS	Soil Description	Remarks	N-Values blows/ft.	Moisture Content %
0 to 2.5 (182.5)	SM	SILTY SAND WITH GRAVEL; medium dense, damp, fine to coarse grained, brown, gravel is fine to coarse	From 0-5': pH = 6.6 SR = 36,000 ohm-cm Cl < 0.0040% SS < 0.0040%	27	1.8
2.5 to 5 (180)	SP	POORLY GRADED SAND; medium dense, damp, fine to medium grained, light brown	DD = 95.0 pcf	20	3.2
		At 3.5 feet - POORLY GRADED SAND WITH GRAVEL; loose, damp, fine to coarse grained, light brown, coarse gravel piece in shoe of sampler (2 inches by 1.5 inches)	From 3.5-5': DD = 87.1 pcf Gravel = 17.2% Sand = 79.0% -200 = 3.8% ø = 35° c = 390 psf	12	3.0
7.5 to 10 (175)		Increase in grain size, increase in fine gravel, decrease in coarse gravel	From 8.5-10': Gravel = 21.6% Sand = 76.8% -200 = 1.6%	6	2.3
10 to 12.5 (172.5)	SP-SM	POORLY GRADED SAND WITH SILT; dense, damp, fine to coarse grained, brown, with some fine to coarse gravel (coarse gravel in cuttings)		32	2.2
12.5 to 13.5 (170)	GP	POORLY GRADED GRAVEL WITH SAND; medium dense, damp, fine to coarse grained, brown, gravel is	From 13.5-14.25': Gravel = 63.4% Sand = 32.0% -200 = 4.6%	18	13.2
	SW-SM				

Notes: ** Groundwater initially encountered during drilling at 18.2 feet. After drilling, groundwater measured at 20.5 feet.*

Figure Number

MOORE TWINING ASSOCIATES, INC.

Test Boring: B-1

Project: Proposed Garland Ranch Regional Park Classroom Project, Carmel Valley

Project Number: H07802.01

Drilled By: J.S.

Drill Type: CME 75

Auger Type: 6-5/8" O.D. Hollow Stem Auger

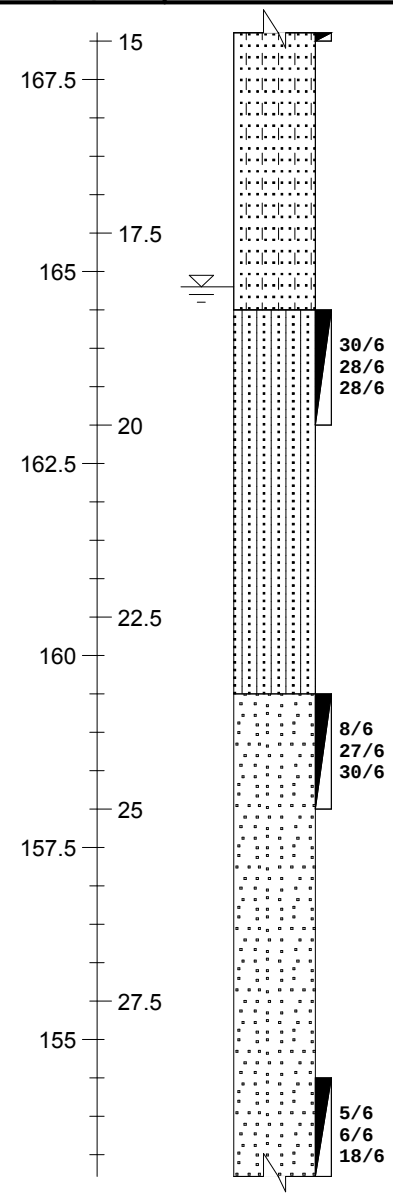
Hammer Type: 140 Pound Auto Trip

Logged By: A.H.

Date: October 19, 2023

Elevation: 183 feet AMSL

Depth to Groundwater First Encountered During Drilling: 18.2 feet*



ELEVATION/ DEPTH (feet)	SOIL SYMBOLS SAMPLER SYMBOLS AND FIELD TEST DATA	USCS	Soil Description	Remarks	N-Values blows/ft.	Moisture Content %
			predominantly coarse gravel	From 14.25-15': Gravel = 20.5% Sand = 73.9% -200 = 5.6%		
			WELL GRADED SAND WITH SILT AND GRAVEL; medium dense, moist, fine to coarse grained, brown, gravel is predominantly fine gravel			
			At 18.2 feet - Wet			
		SM	SILTY SAND; very dense, wet, fine to coarse grained, brown, with some fine gravel, scattered cobbles in cuttings		56	13.5
		SP	POORLY GRADED SAND; very dense, wet, fine to coarse grained, brown, with some fine gravel, scattered cobbles in cuttings		57	16.1
			Medium dense		24	

Notes: ** Groundwater initially encountered during drilling at 18.2 feet. After drilling, groundwater measured at 20.5 feet.*

Figure Number

MOORE TWINING ASSOCIATES, INC.

Test Boring: B-1

Project: Proposed Garland Ranch Regional Park Classroom Project, Carmel Valley

Project Number: H07802.01

Drilled By: J.S.

Drill Type: CME 75

Auger Type: 6-5/8" O.D. Hollow Stem Auger

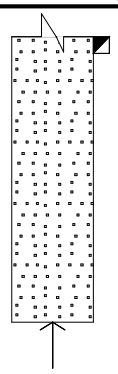
Hammer Type: 140 Pound Auto Trip

Logged By: A.H.

Date: October 19, 2023

Elevation: 183 feet AMSL

Depth to Groundwater First Encountered During Drilling: 18.2 feet*

ELEVATION/ DEPTH (feet)	SOIL SYMBOLS SAMPLER SYMBOLS AND FIELD TEST DATA	USCS	Soil Description	Remarks	N-Values blows/ft.	Moisture Content %
30 / 152.5 / 32.5 / 150 / 35 / 147.5 / 37.5 / 145 / 40 / 142.5 / 42.5 / 140			Bottom of Boring B-1 (Boring terminated at 33.5 feet due to 3.5 feet of heave up inside hollow stem augers, not allowing sample to be collected)			

Notes: ** Groundwater initially encountered during drilling at 18.2 feet. After drilling, groundwater measured at 20.5 feet.*

Figure Number

MOORE TWINING ASSOCIATES, INC.

Test Boring: B-2

Project: Proposed Garland Ranch Regional Park Classroom Project, Carmel Valley

Project Number: H07802.01

Drilled By: J.S.

Drill Type: CME 75

Auger Type: 6-5/8" O.D. Hollow Stem Auger

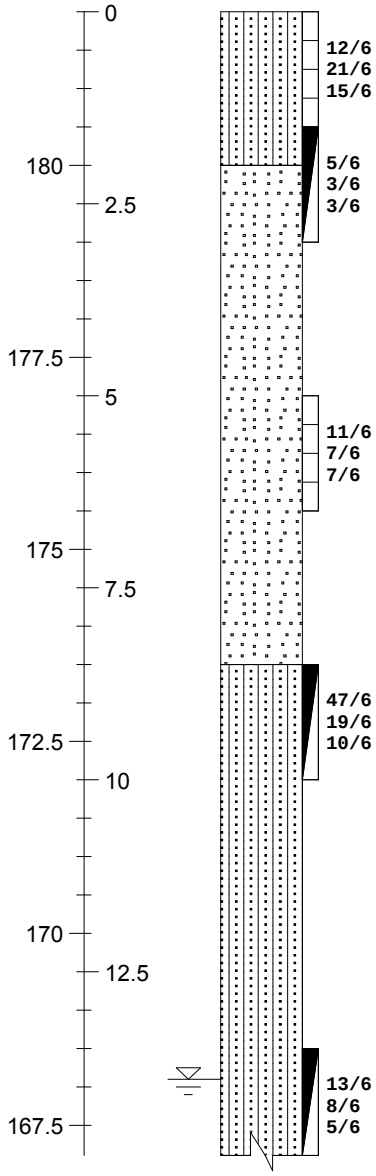
Hammer Type: 140 Pound Auto Trip

Logged By: A.H.

Date: October 19, 2023

Elevation: 182 feet AMSL

Depth to Groundwater First Encountered During Drilling: 13.9 feet*



ELEVATION/ DEPTH (feet)	SOIL SYMBOLS SAMPLER SYMBOLS AND FIELD TEST DATA	USCS	Soil Description	Remarks	N-Values blows/ft.	Moisture Content %
		SM	SILTY SAND; medium dense, damp, fine to coarse grained, brown, with a little fine to coarse gravel	Ring sample disturbed	36	1.3
			At 1.5 feet - Loose		6	
		SP	POORLY GRADED SAND; loose, damp, fine to medium grained, brown, trace fine gravel	From 2-3': Gravel = 1.8% Sand = 93.3% -200 = 4.9%		2.4
			At 5 feet - POORLY GRADED SAND WITH GRAVEL; loose, moist, fine to medium grained, gravel is predominantly course gravel	From 5-6.5': DD = 95.0 pcf Gravel = 18.7% Sand = 78.5% -200 = 2.8%	14	5.9
			At 6 feet - Scattered cobbles in cuttings			
		SM	SILTY SAND; medium dense, damp, predominantly fine grained, brown, with a little medium to coarse sand and fine gravel		29	
			At 12.5 feet - Scattered cobbles in cuttings			
			At 13.9 feet - Wet	No recovery	13	

Notes: ** Groundwater initially encountered during drilling at 13.9 feet. After drilling, groundwater measured at 12.9 feet.*

Figure Number

MOORE TWINING ASSOCIATES, INC.

Test Boring: B-2

Project: Proposed Garland Ranch Regional Park Classroom Project, Carmel Valley

Project Number: H07802.01

Drilled By: J.S.

Drill Type: CME 75

Auger Type: 6-5/8" O.D. Hollow Stem Auger

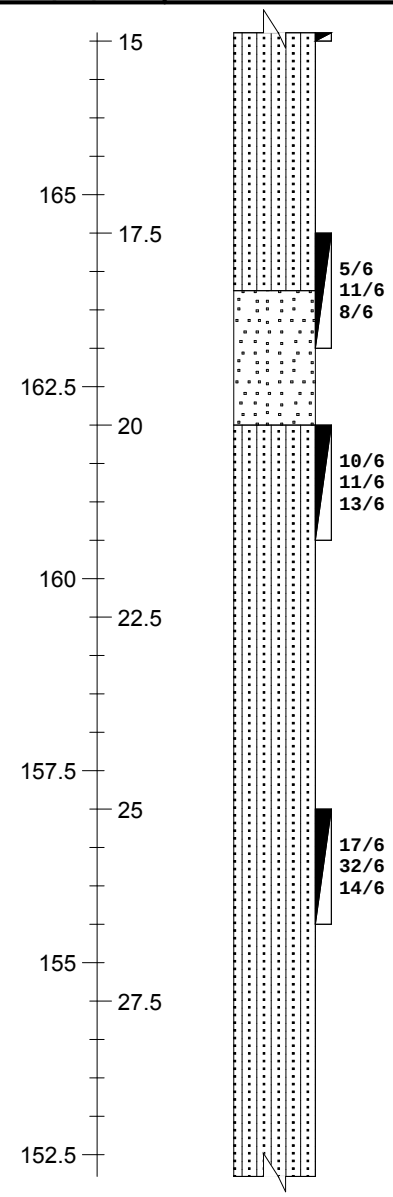
Hammer Type: 140 Pound Auto Trip

Logged By: A.H.

Date: October 19, 2023

Elevation: 182 feet AMSL

Depth to Groundwater First Encountered During Drilling: 13.9 feet*



ELEVATION/ DEPTH (feet)	SOIL SYMBOLS SAMPLER SYMBOLS AND FIELD TEST DATA	USCS	Soil Description	Remarks	N-Values blows/ft.	Moisture Content %
17.5	5/6 11/6 8/6	SP	Medium dense, fine to medium grained, brown to reddish brown, trace fine gravel	From 17.5-18.25': Gravel = 3.3% Sand = 80.2% -200 = 16.5%	19	27.6
						15.1
20	10/6 11/6 13/6	SM	POORLY GRADED SAND WITH GRAVEL; medium dense, wet, fine to coarse grained, grayish brown, gravel is fine to coarse		24	19.9
			At 20 feet - SILTY SAND; medium dense, wet, fine to medium grained, brown, with a little fine gravel			
25	17/6 32/6 14/6		SILTY SAND WITH GRAVEL; dense, wet, fine to coarse grained, grayish brown, gravel is fine to coarse		46	
27.5						

Notes: ** Groundwater initially encountered during drilling at 13.9 feet. After drilling, groundwater measured at 12.9 feet.*

Figure Number

MOORE TWINING ASSOCIATES, INC.

Test Boring: B-2

Project: Proposed Garland Ranch Regional Park Classroom Project, Carmel Valley

Project Number: H07802.01

Drilled By: J.S.

Drill Type: CME 75

Auger Type: 6-5/8" O.D. Hollow Stem Auger

Hammer Type: 140 Pound Auto Trip

Logged By: A.H.

Date: October 19, 2023

Elevation: 182 feet AMSL

Depth to Groundwater First Encountered During Drilling: 13.9 feet*

ELEVATION/ DEPTH (feet)	SOIL SYMBOLS SAMPLER SYMBOLS AND FIELD TEST DATA	USCS	Soil Description	Remarks	N-Values blows/ft.	Moisture Content %
30	12/6 7/6 9/6	SP-SM	POORLY GRADED SAND WITH SILT AND GRAVEL; medium dense, wet, fine to coarse grained, grayish brown, gravel is fine to coarse	From 30-31.5': Gravel = 17.0% Sand = 74.3% -200 = 8.7%	16	18.1
150 32.5			Bottom of boring B-2 at 32 feet (Boring terminated due to auger refusal on gravelly/cobbly soils and due to 3.5 feet of heave up inside augers, not allowing sample to be collected)			
147.5 35						
145 37.5						
142.5 40						
140 42.5						
137.5						

Notes: ** Groundwater initially encountered during drilling at 13.9 feet. After drilling, groundwater measured at 12.9 feet.*

Figure Number

KEY TO SYMBOLS

Symbol	Description
Strata symbols	
	Silty Sand
	Poorly Graded Sand
	Poorly Graded Sand with Silt
	Poorly graded gravel
	Well graded sand with silt
Misc. Symbols	
	Drill rejection

Symbol	Description
	Boring continues
	Water table during drilling
Soil Samplers	
	Standard penetration test
	California Modified split barrel ring sampler

Notes:

1. Exploratory borings were drilled on 10/19/23 using a CME 75 drill rig equipped with 6-5/8" outside diameter hollow stem augers.

2. Groundwater was encountered in both borings, see logs.

3. Boring locations were measured or paced from existing features.

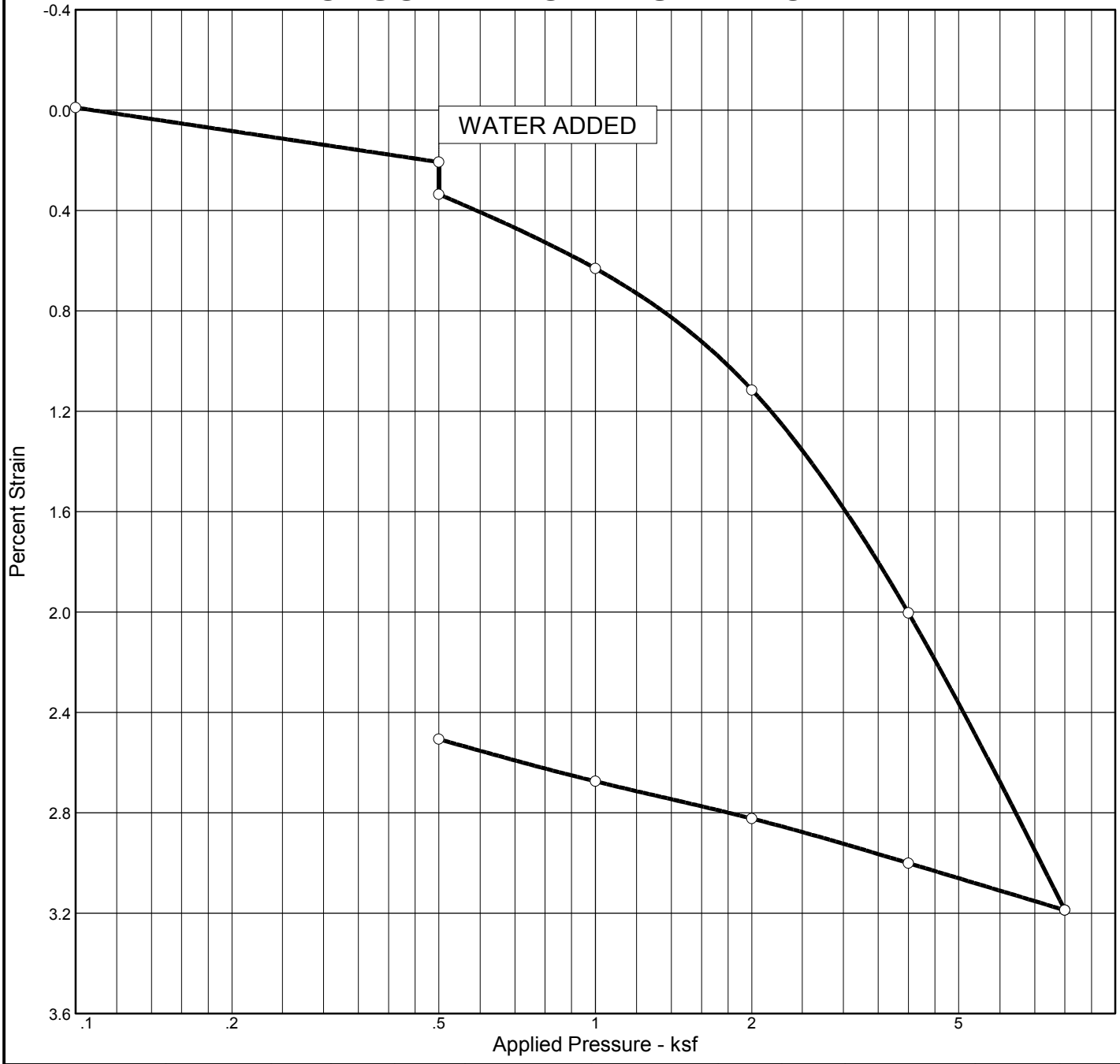
4. These logs are subject to the limitations, conclusions, and recommendations in this report.

5. The "N-value" reported for the California Modified Split Barrel Sampler is the uncorrected field blow count. This value should not be interpreted as an SPT equivalent N-value.

6. Results of tests conducted on samples recovered are reported on the logs.

DD = Natural dry density (pcf)	LL = Liquid Limit (%)
+4 = Percent retained on the No. 4 sieve(%)	PI = Plasticity Index (%)
-200 = Percent passing the No. 200 sieve (%)	EI = Expansion Index
Sand = Percent passing the No. 4 sieve and retained on No. 200 sieve (%)	Gravel = Percent passing 3-inch & retained on No. 4 sieves(%)
pH = Soil pH	SR = Soil resistivity (ohms-cm)
SS = Soluble sulfates (%)	Cl = Soluble chlorides (%)
ø = Internal Angle of Friction (degrees)	c = Cohesion (psf)
pcf = Pounds per cubic foot	psf = Pounds per square foot
O.D. = Outside diameter	AMSL = Above mean sea level
N/A = Not applicable	N/E = Not encountered

CONSOLIDATION TEST REPORT



Natural		Dry Dens. (pcf)	LL	PI	Sp. Gr.	Overburden (ksf)	P_c (ksf)	C_c	C_s	Swell Press. (ksf)	Clpse. %	e_o
Sat.	Moist.											
20.3 %	5.3 %	98.1			2.65		2.59	0.07	0.01		0.1	0.686

MATERIAL DESCRIPTION	USCS	AASHTO
Poorly graded sand	SP	

Project No. H07802.01 **Client:** Monterey Peninsula Regional Park District

Project: Proposed Classroom Project, Carmel Valley

Source: **Sample No.:** B-1 **Elev./Depth:** 2-3.5'

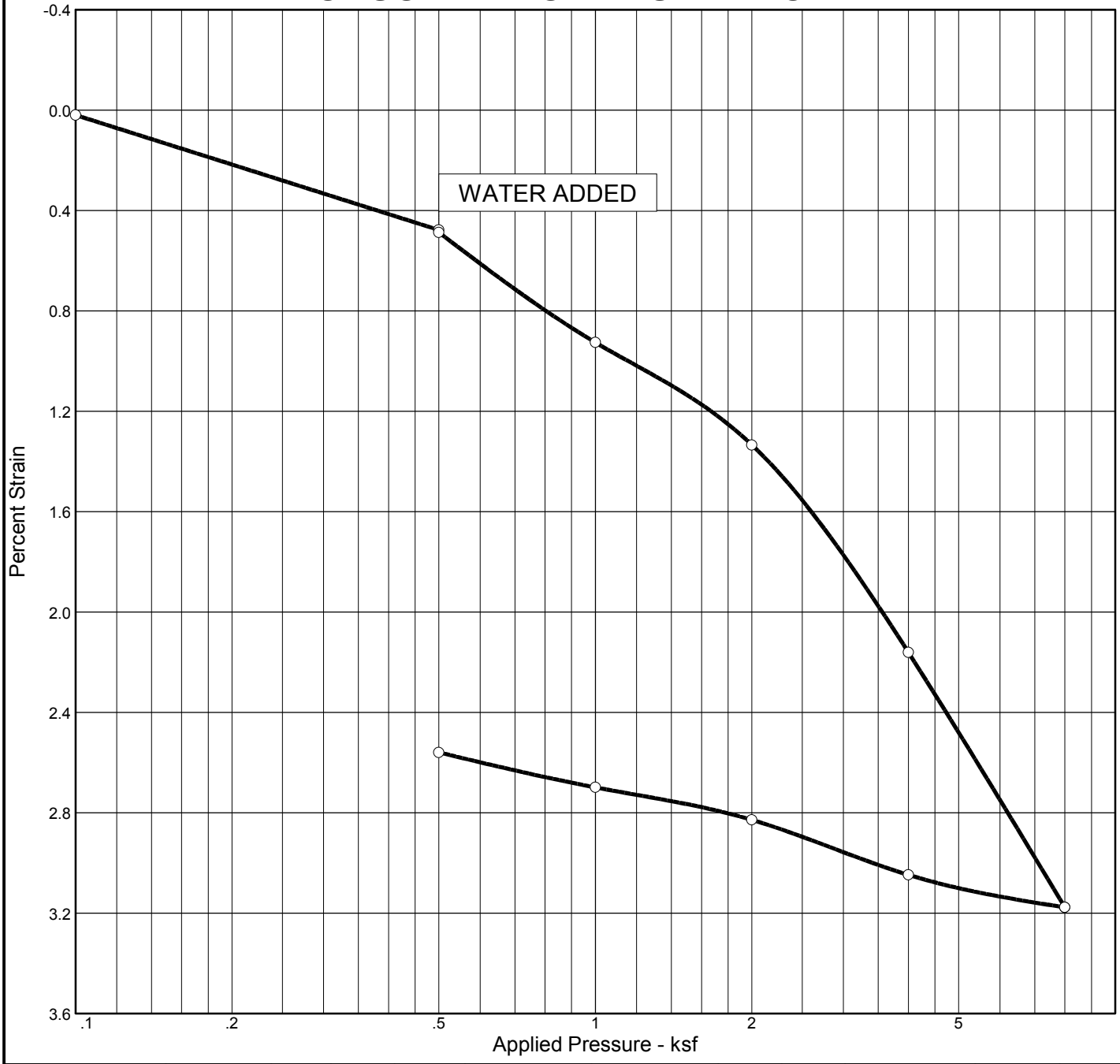
Remarks:

Moore Twining Associates, Inc.

Fresno, CA

Figure

CONSOLIDATION TEST REPORT



Natural		Dry Dens. (pcf)	LL	PI	Sp. Gr.	Overburden (ksf)	P_c (ksf)	C_c	C_s	Swell Press. (ksf)	Swell %	e_o
Sat.	Moist.											
12.7 %	3.3 %	98.3			2.65		2.43	0.06	0.01			0.682

MATERIAL DESCRIPTION	USCS	AASHTO
Poorly graded sand with gravel	SP	

Project No. H07802.01 **Client:** Monterey Peninsula Regional Park District

Project: Proposed Classroom Project, Carmel Valley

Source: **Sample No.:** B-1 **Elev./Depth:** 3.5-5'

Moore Twining Associates, Inc.

Fresno, CA

Remarks:

Figure